The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 September 2007.

INCH-POUND

MIL-PRF-19500/555J
8 June 2007
SUPERSEDING
MIL-PRF-19500/555H
8 December 2003

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR, N-CHANNEL, SILICON,
TYPES 2N6788, 2N6788U, 2N6790, 2N6790U, 2N6792, 2N6792U, 2N6794, AND 2N6794U,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for an N-channel, enhancement-mode, MOSFET, power transistor. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figures 1 (TO-205AF), 2 (LCC), and figures 3, 4, 5, and 6 for JANHC and JANKC (die) dimensions.

* 1.3 Maximum ratings. (Unless otherwise specified, T_A = +25°C).

Type	P_T T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	V_{DS} and V_{DG} 70,000 ft. altitude
	W	V dc	V dc	V dc	
2N6788, 2N6788U	0.8	100	100	±20	
2N6790, 2N6790U	0.8	200	200	±20	
2N6792, 2N6792U	0.8	400	400	±20	300
2N6794, 2N6794U	0.8	500	500	±20	300

Type	P_T (1) T_C = +25°C	$R_{\theta JC}$ (2)	I_{D1} (3) (4) T_C = +25°C	I_{D2} T_C = +100°C	I_S	I_{DM} (5)	T_J and T_{STG}
	W	°C/W	A dc	A dc	A dc	A (pk)	°C
2N6788	20	6.25	6.0	3.5	6.0	24	-55°
2N6790	20	6.25	3.5	2.25	3.5	14	to
2N6792	20	6.25	2.0	1.25	2.0	8	+150°
2N6794	20	6.25	1.5	1.0	1.5	6	
2N6788U	14	8.93	4.5	2.8	4.5	18	
2N6790U	14	8.93	2.8	1.8	2.8	11	
2N6792U	14	8.93	1.8	1.13	1.8	7.2	
2N6794U	14	8.93	1.4	0.88	1.4	5.6	

See notes on next page.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

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* 1.3 Maximum ratings. Continued.

(1) Derate linearly 0.16 W/°C for T_C > +25°C for "non-U" suffix versions, 0.11 W/°C for T_C > +25°C for "U" suffix versions.
(2) See figure 7, thermal impedance curves.
(3) The following formula derives the maximum theoretical I_D limit. I_D is also limited by package and internal wires and may be limited due to pin diameter.

$$I_D = \sqrt{\frac{T_J(\max) - T_C}{R_{\theta JC}\, x (R_{DS(on)}\, at\, T_J(\max)}}$$

(4) See figure 8, maximum drain current graph.
(5) I_{DM} = 4 I_{D1}; I_{D1} as calculated in footnote (3).

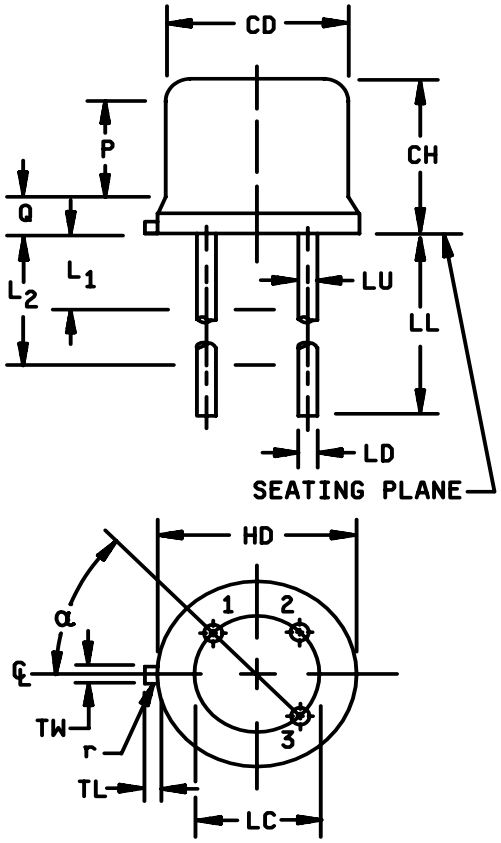
1.4 Unless otherwise specified, primary electrical characteristics at T_C = +25°C.

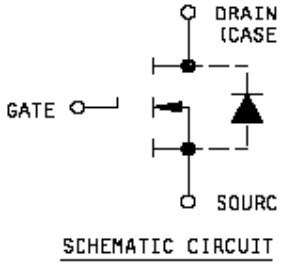
Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		Max I_{DSS1} V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)}$ (1) V_{GS} = 10 V dc $I_D = I_{D2}$	
					T_J = +25°C	T_J = +150°C
	V dc	V dc Min	Max	µA dc	Ohms	Ohms
2N6788, 2N6788U	100	2.0	4.0	25	0.30	0.60
2N6790, 2N6790U	200	2.0	4.0	25	0.80	1.80
2N6792, 2N6792U	400	2.0	4.0	25	1.80	4.50
2N6794, 2N6794U	500	2.0	4.0	25	3.00	7.50

(1) Pulsed (see 4.5.1).

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.160	.180	4.07	4.57	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8,12
LU	.016	.019	0.41	0.48	7,8
L1		.050		1.27	7,8
L2	.250		6.35		7,8
P	.100		2.54		
Q		.050		1.27	5
TL	.029	.045	0.74	1.14	3,4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45° TP		45° TP		6





SCHEMATIC CIRCUIT

NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TL shall be held for a minimum length of .011 inch (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = source, lead 2 = gate, lead 3 = drain.

FIGURE 1. Physical dimensions (similar to TO-205AF) 2N6788, 2N6790, 2N6792, and 2N6794.

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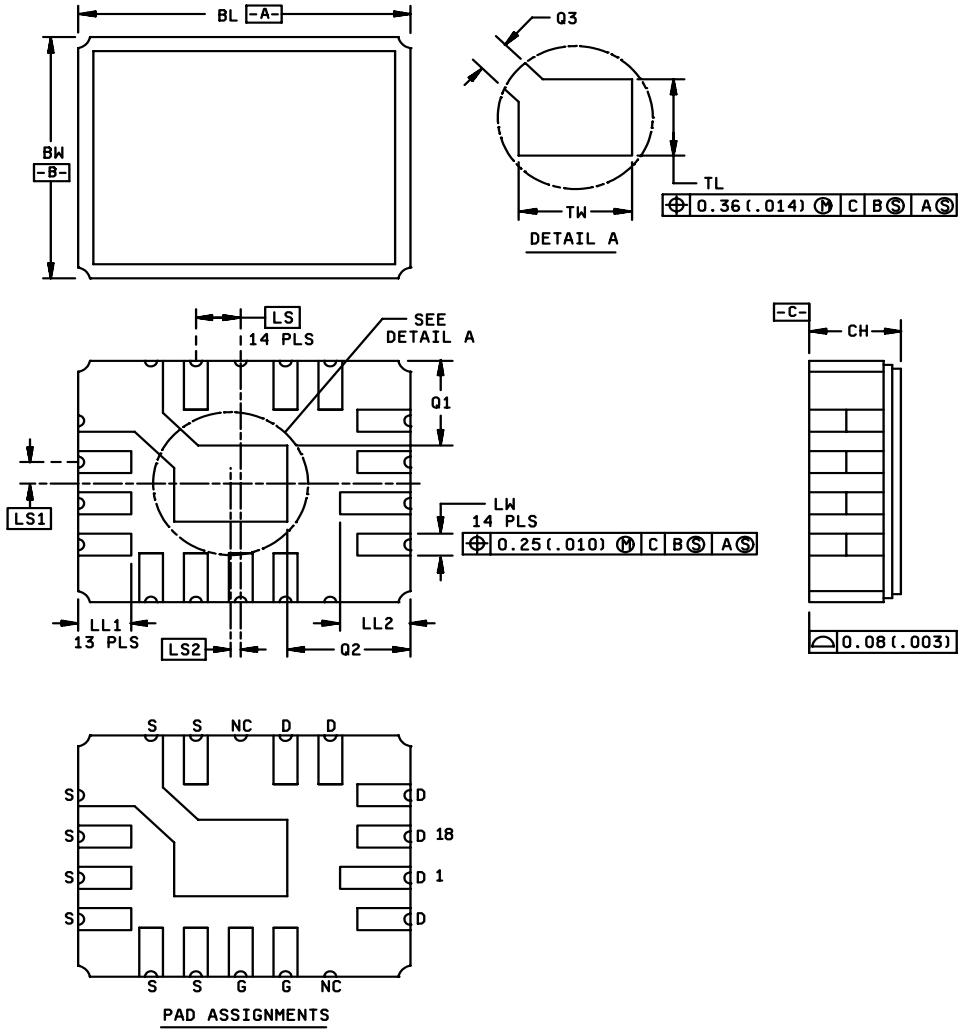


FIGURE 2. Physical dimensions for LCC (2N6788U, 2N6790U, 2N6792U, and 2N6794U).

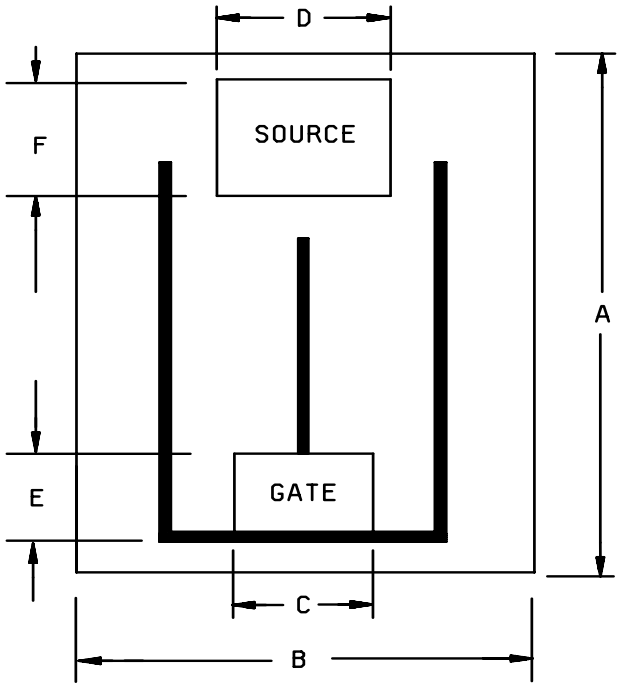
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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.345	.360	8.77	9.14
BW	.280	.295	7.12	7.49
CH	.095	.115	2.42	2.92
LL_1	.040	.055	1.02	1.39
LL_2	.055	.065	1.40	1.65
LS	.050 BSC		1.27 BSC	
LS_1	.025 BSC		0.635 BSC	
LS_2	.008 BSC		0.203 BSC	
LW	.020	.030	0.51	0.76
Q_1	.105 REF		2.67 REF	
Q_2	.120 REF		3.05 REF	
Q_3	.045	.055	1.15	1.39
TL	.070	.080	1.78	2.03
TW	.120	.130	3.05	3.30

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimensions and tolerancing shall be in accordance with ASME Y14.5M.

FIGURE 2. Physical dimensions for LCC (2N6788U, 2N6790U, 2N6792U, and 2N6794U) - Continued.

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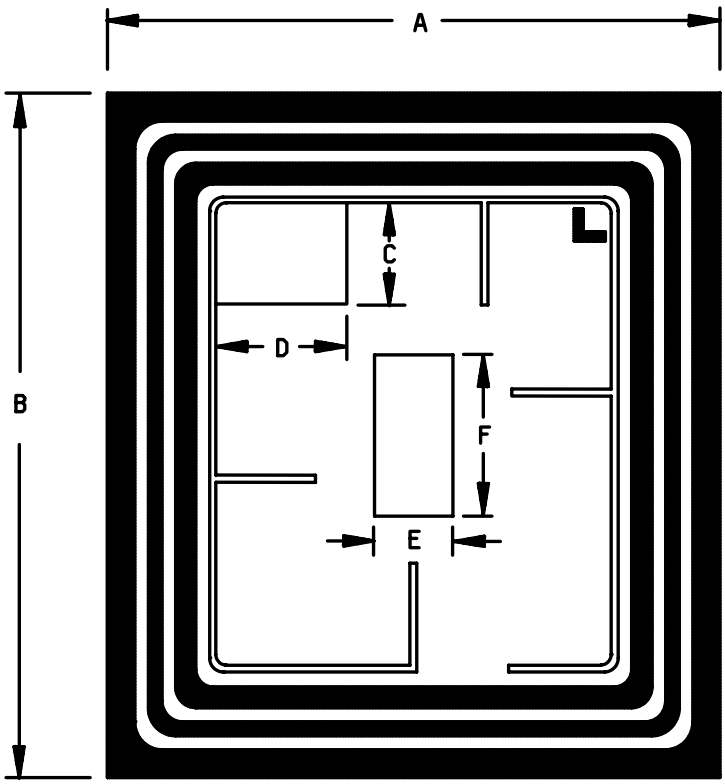
A version

Ltr	Dimensions - 2N6788				Dimensions - 2N6790				Dimensions - 2N6792, 2N6794			
	Inches		Millimeters		Inches		Millimeters		Inches		Millimeters	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
A	.107	.121	2.72	3.07	.094	.112	2.39	2.85	.131	.147	3.33	3.73
B	.078	.088	1.98	2.24	.083	.099	2.11	2.52	.090	.106	2.29	2.69
C	.020	.030	0.51	0.76	.018	.028	0.46	0.71	.022	.032	0.56	0.81
D	.027	.037	0.69	0.94	.028	.038	0.71	0.97	.028	.038	0.71	0.97
E	.013	.023	0.33	0.58	.015	.025	0.38	0.64	.015	.025	0.38	0.64
F	.019	.029	0.48	0.74	.018	.028	0.46	0.71	.020	.030	0.51	0.76

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Die thickness = .019 ± .005 inch (0.48 ± 0.13 mm).
4. Back metal: Cr - Ni - Ag.
5. Top metal: Al.
6. Back contact: Drain.
7. Layout of gate fingers shown is typical, specific layout register in accordance with DSCC Form 36D.
8. See 6.5.

FIGURE 3. Physical dimensions JANHCA and JANKCA.

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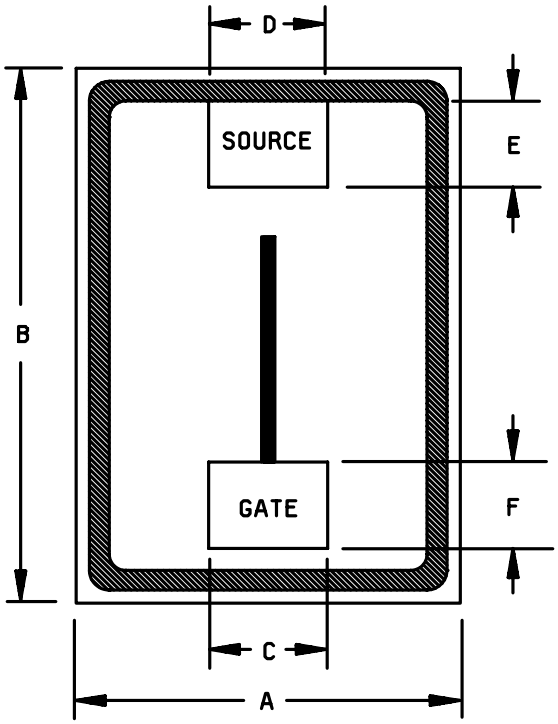
B version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.114	.118	2.9	3.0
B	.120	.124	3.0	3.1
C	.018	.022	0.46	0.56
D	.028	.032	0.71	0.81
E	.018	.022	0.46	0.56
F	.029	.033	0.74	0.84

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Die thickness = .014 ±.005 inch (0.36 ±0.13 mm).
4. Back metal: Al - Ti - Ni.
5. Top metal: Al.
6. Back contact: Drain.

FIGURE 4. Physical dimensions JANHCB and JANKCB.

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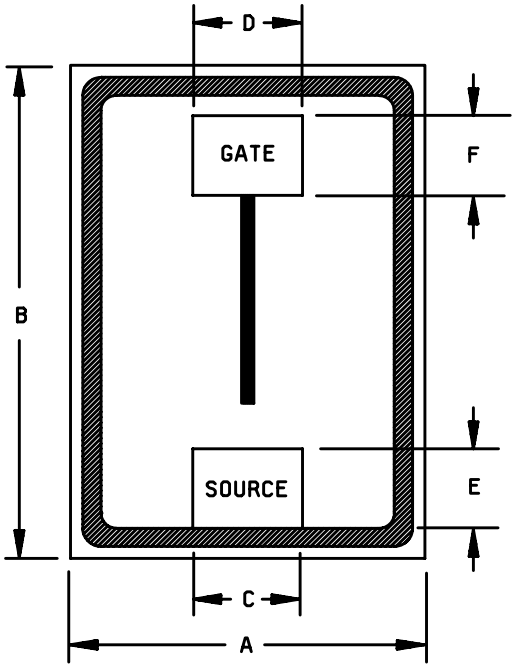
C version

Ltr	Dimensions 2N6788, 2N6790			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.0858	.0898	2.18	2.28
B	.087	.091	2.21	2.31
C	.0258	.0298	0.65	0.76
D	.0253	.0293	0.64	0.74
E	.017	.021	0.43	0.53
F	.016	.020	0.41	0.51

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Die thickness = .015 ±.005 inch (0.38 ±0.13 mm).
4. Back metal: Ag - Ti - Ni.
5. Top metal: Al.
6. Back contact: Drain.

FIGURE 5. Physical dimensions JANHCC and JANKCC.

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D version

Ltr	Dimensions 2N6792, 2N6794			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.093	.102	2.36	2.59
B	.140	.144	3.56	3.66
C	.025	.029	0.64	0.74
D	.026	.030	0.66	0.76
E	.016	.020	0.41	0.51
F	.017	.021	0.43	0.53

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Die thickness = .015 ±.005 inch (0.38 ±0.13 mm).
4. Back metal: Ag - Ti - Ni.
5. Top metal: Al.
6. Back contact: Drain.
8. See 6.5.

FIGURE 6. Physical dimensions JANHCD and JANKCD.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

I_{AS} - Rated avalanche current, nonrepetitive.
nC - nano coulomb.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 (TO-205AF), 2 (LCC), and figures 3, 4, 5, and 6 for JANHC and JANKC (die) herein.

3.4.1 Lead material and finish. Lead material shall be Kovar or Alloy 52; a copper core is permitted (for TO-205AF). Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

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3.6 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.6.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.6).

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care should be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. $R \leq 100\ k\Omega$, whenever bias voltage is to be applied drain to source.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

* c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with MIL-PRF-19500.

4.2.1 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be as specified in MIL-PRF-19500.

* 4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed by the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500) (1) (2)	Measurement	
	JANS level	JANTX and JANTXV levels
(3)	Gate stress test (see 4.3.2)	Gate stress test (see 4.3.2)
(3) (4)	Method 3470 of MIL-STD-750 (see 4.3.3) optional	Method 3470 of MIL-STD-750 (see 4.3.3) optional
(3) 3c	Method 3161 of MIL-STD-750 (see 4.3.4)	Method 3161 of MIL-STD-750 (see 4.3.4)
9	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, subgroup 2 of table I herein	Not applicable
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	Subgroup 2 of table I herein; I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$,$V_{GS(TH)1}$, ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater.	Subgroup 2 of table I herein, I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$,$V_{GS(TH)1}$
12	Method 1042 of MIL-STD-750, test condition A	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$= ±20 percent of initial value.	Subgroup 2 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$= ±20 percent of initial value.

(1) At the end of the test program, I_{GSSF1} I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1}, and $V_{GS(th)1}$ shall be invoked.
* (3) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.
(4) Method 3470 is optional if performed as a sample in group A, subgroup 5.

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4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-PRF-19500, as a minimum, die shall be 100 percent probed in accordance with table I, subgroup 2, except test current shall not exceed 20 A.

4.3.2 Gate stress test. Apply V_{GS} = +30 V minimum for t = 250 µs minimum.

4.3.3 Single pulsed unclamped inductive switching.

a. Peak current, I_D ..... 2.2 A.

b. Peak gate voltage, V_{GS} ..... 10 V.

c. Gate to source resistor, R_{GS} ..... $25 \leq R_g \leq 200\Omega$.

d. Initial case temperature ..... +25°C, +10°C, -5°C.

e. Inductance, L ..... 100 µH ±10 percent.

f. Number of pulses to be applied ..... 1 pulse.

* 4.3.4 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). Measurement delay time (t_{MD}) = 70 µs max. See table II, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for conformance inspection in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the inspections of table I, subgroup 2 herein.

4.4.2.1 Group B inspection table E-VIA (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1051	Test condition G.
B3	2037	Test condition A. All internal wires for each device shall be pulled separately. If group B3 is to be continued to C6, strength test may be performed after C6.
B4	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle.
B5	1042	A separate sample may be pulled for each test. Accelerated steady-state reverse bias; test condition A, V_{DS} = rated, T_A = +175°C, t = 120 hours, read and record $V_{BR(DSS)}$ (pre and post) at I_D = -1 mA. Read and record I_{DSS} (pre and post).
B5	1042	Accelerated steady-state gate stress; test condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.

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4.4.2.2 Group B inspection table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	1051	Test condition G.
B3	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle.
B3	2037	Test condition A. All internal wires for each device shall be pulled separately. If group B3 is to be continued to C6, bond strength test may be performed after C6.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the inspections of table I, subgroup 2 herein.

4.4.3.1 Group C inspection (table E-VII of MIL-PRF-19500).

	Subgroup	Method	Conditions
	C2	2036	Test condition E (not required for LCC).
*	C5	3161	See 4.3.4, for $R_{\theta JC}$ = 6.25°C/W for 2N6788, 2N6790, 2N6792, and 2N6794 $R_{\theta JC}$ = 8.93°C/W for 2N6788U, 2N6790U, 2N6792U, and 2N6794U
	C6	1042	Test condition D, 6,000 cycles. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
	Subgroup 1						
	Visual and mechanical inspection	2071					
	Subgroup 2						
*	Thermal impedance 2/	3161	See 4.3.4	$Z_{\theta JC}$			°C/W
	Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = 1 mA dc, condition C	$V_{(BR)DSS}$			
	2N6788, 2N6788U				100		V dc
	2N6790, 2N6790U				200		V dc
	2N6792, 2N6792U				400		V dc
	2N6794, 2N6794U				500		V dc
	Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA dc	$V_{GS(th)1}$	2.0	4.0	V dc
*	Gate current	3411	V_{GS} = + 20 V dc, V_{DS} = 0, bias condition C	I_{GSSF1}		+100	nA dc
*	Gate current	3411	V_{GS} = - 20 V dc, V_{DS} = 0, bias condition C	I_{GSSR1}		- 100	nA dc
	Drain current	3413	V_{GS} = 0 V dc, V_{DS} = 80 percent of rated V_{DS}; bias condition C	I_{DSS1}		25	µA dc
	Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1); condition A, I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			
	2N6788, 2N6788U					0.30	Ω
	2N6790, 2N6790U					0.8	Ω
	2N6792, 2N6792U					1.8	Ω
	2N6794, 2N6794U					3.0	Ω
	Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1); condition A, I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			
	2N6788, 2N6788U					.35	Ω
	2N6790, 2N6790U					.85	Ω
	2N6792, 2N6792U					1.90	Ω
	2N6794, 2N6794U					3.10	Ω
	Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1), I_S = I_{D1}	V_{SD}			
	2N6788, 2N6788U					1.8	V
	2N6790, 2N6790U					1.5	V
	2N6792, 2N6792U					1.4	V
	2N6794, 2N6794U					1.2	V

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate to source voltage threshold	3403	$V_{DS} \geq V_{GS}$, $I_D = 0.25$ mA dc	$V_{GS(th)2}$	1.0		V dc
Gate current	3411	Bias condition C, $V_{GS} = \pm 20$ V dc, $V_{DS} = 0$ V dc	I_{GSS2}		± 200	nA dc
Drain current	3413	Bias condition C, $V_{GS} = 0$ V dc, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		0.25	mA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1); I_D = rated I_{D2}	$r_{DS(on)3}$			
2N6788, 2N6788U					0.54	Ω
2N6790, 2N6790U					1.50	Ω
2N6792, 2N6792U					4.00	Ω
2N6794, 2N6794U					6.60	Ω
Low temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage threshold	3403	$V_{DS} \geq V_{GS}$, $I_D = 0.25$ mA dc	$V_{GS(th)3}$		5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1}, V_{GS} = 10 V dc, gate drive impedance = 7.5Ω				
Turn-on delay time			$t_{d(on)}$			
2N6788, 2N6788U		V_{DD} = 35 V dc			40	ns
2N6790, 2N6790U		V_{DD} = 74 V dc			40	ns
2N6792, 2N6792U		V_{DD} = 175 V dc			40	ns
2N6794, 2N6794U		V_{DD} = 225 V dc			40	ns
Rise time			t_r			
2N6788, 2N6788U		V_{DD} = 35 V dc			70	ns
2N6790, 2N6790U		V_{DD} = 74 V dc			50	ns
2N6792, 2N6792U		V_{DD} = 175 V dc			35	ns
2N6794, 2N6794U		V_{DD} = 225 V dc			30	ns
Turn-off delay time			$t_{d(off)}$			
2N6788, 2N6788U		V_{DD} = 35 V dc			40	ns
2N6790, 2N6790U		V_{DD} = 74 V dc			50	ns
2N6792, 2N6792U		V_{DD} = 175 V dc			60	ns
2N6794, 2N6794U		V_{DD} = 225 V dc			60	ns
Fall time			t_f			
2N6788, 2N6788U		V_{DD} = 35 V dc			70	ns
2N6790, 2N6790U		V_{DD} = 74 V dc			50	ns
2N6792, 2N6792U		V_{DD} = 175 V dc			35	ns
2N6794, 2N6794U		V_{DD} = 225 V dc			30	ns

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
	Subgroup 5						
	Safe operating area test	3474	See figure 9; V_{DS} = 80 percent of rated $V_{BR(DSS)}$, $V_{DS} \leq 200$ V; t_p = 10 ms				
	Electrical measurements		See table I, subgroup 2				
	Single pulse unclamped inductive switching 3/	3470	See 4.3.3, c = 0, 116 devices				
	Electrical measurements		See table I, subgroup 2				
	Subgroup 6						
	Not applicable						
	Subgroup 7						
	Gate charge	3471	Condition B				
	On-state charge			$Q_{g(on)}$			
*	2N6788, 2N6788U					18.0	nC
	2N6790, 2N6790U					14.3	nC
*	2N6792, 2N6792U					22.0	nC
*	2N6794, 2N6794U					25.0	nC
	Gate to source charge			Q_{gs}			
	2N6788, 2N6788U					4.0	nC
	2N6790, 2N6790U					3.0	nC
*	2N6792, 2N6792U					3.0	nC
*	2N6794, 2N6794U					6.0	nC
	Gate to drain charge			Q_{gd}			
*	2N6788, 2N6788U					9.0	nC
	2N6790, 2N6790U					9.0	nC
*	2N6792, 2N6792U					14.0	nC
*	2N6794, 2N6794U					18.0	nC
	Reverse recovery time	3473	di/dt = 100 A/µs, $V_{DD} \leq 50$ V, I_D = I_{D1}	t_{rr}			
	2N6788, 2N6788U					240	ns
	2N6790, 2N6790U					400	ns
	2N6792, 2N6792U					650	ns
	2N6794, 2N6794U					900	ns

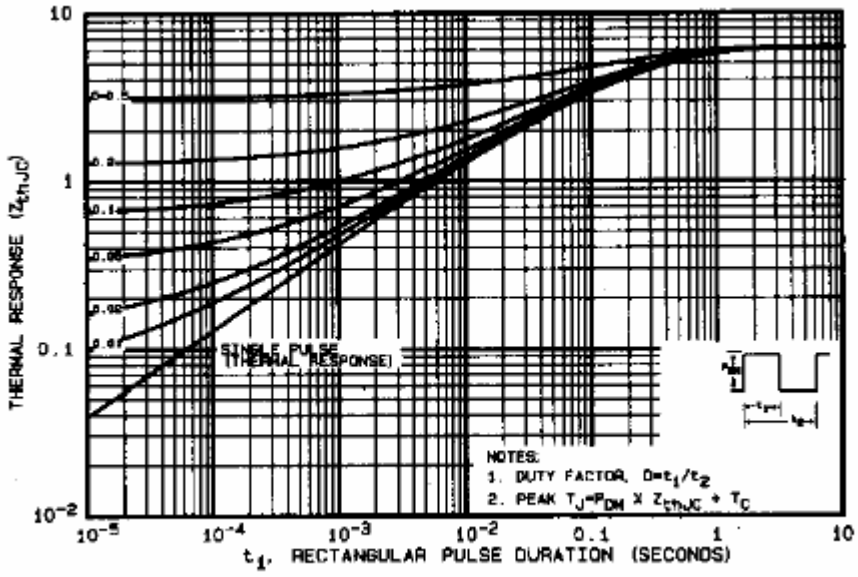
1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.
3/ This test need not be performed in group A when performed as a 100-percent screen.

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TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycle	1051	Condition G, 200 cycles	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2	
Subgroup 2			45 devices c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Steady-state gate bias	1042	Condition B, 1,000 hours, V_{GS} = 80 percent of rated (see 1.3)	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			3 devices c = 0
Barometric pressure (reduced)	1001	Test condition C, $V_{ISO} = V_{DS}$, $I_{(ISO)}$ = .25 mA (max)	
2N6792, 2N6792U 2N6794, 2N6794U		V_{DS} = 300 V V_{DS} = 300 V	
Subgroup 6			3 devices
ESD	1020	Not required for devices classified as ESD class 1.	
Subgroup 10			
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer	22 devices c = 0

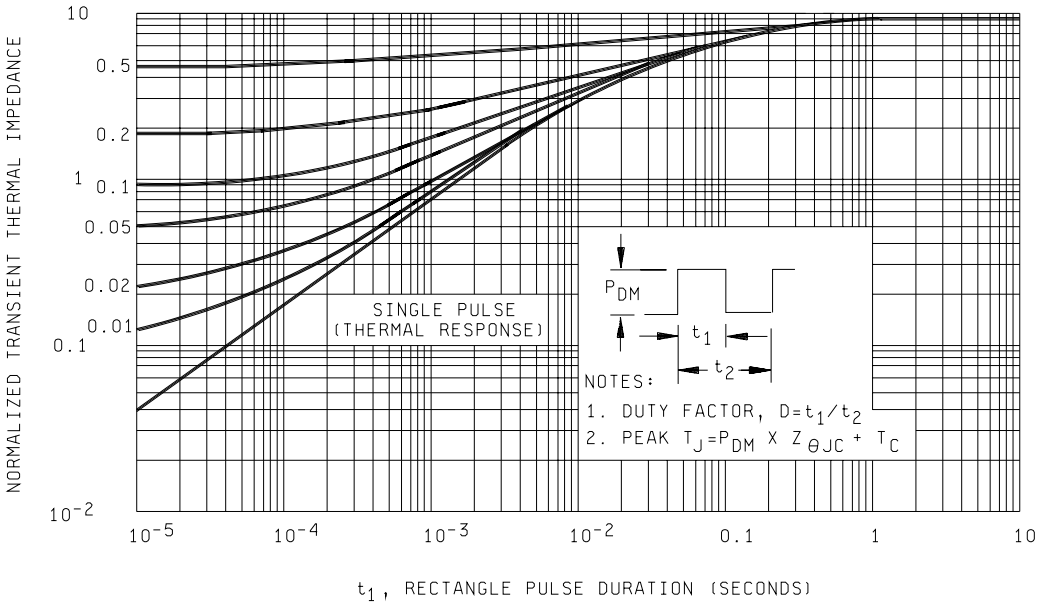
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2N6788, 2N6790, 2N6792, and 2N6794 only

* FIGURE 7. Thermal impedance curves.

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2N6788U, 2N6790U, 2N6792U, and 2N6794U only

* FIGURE 7. Thermal impedance curves- Continued.

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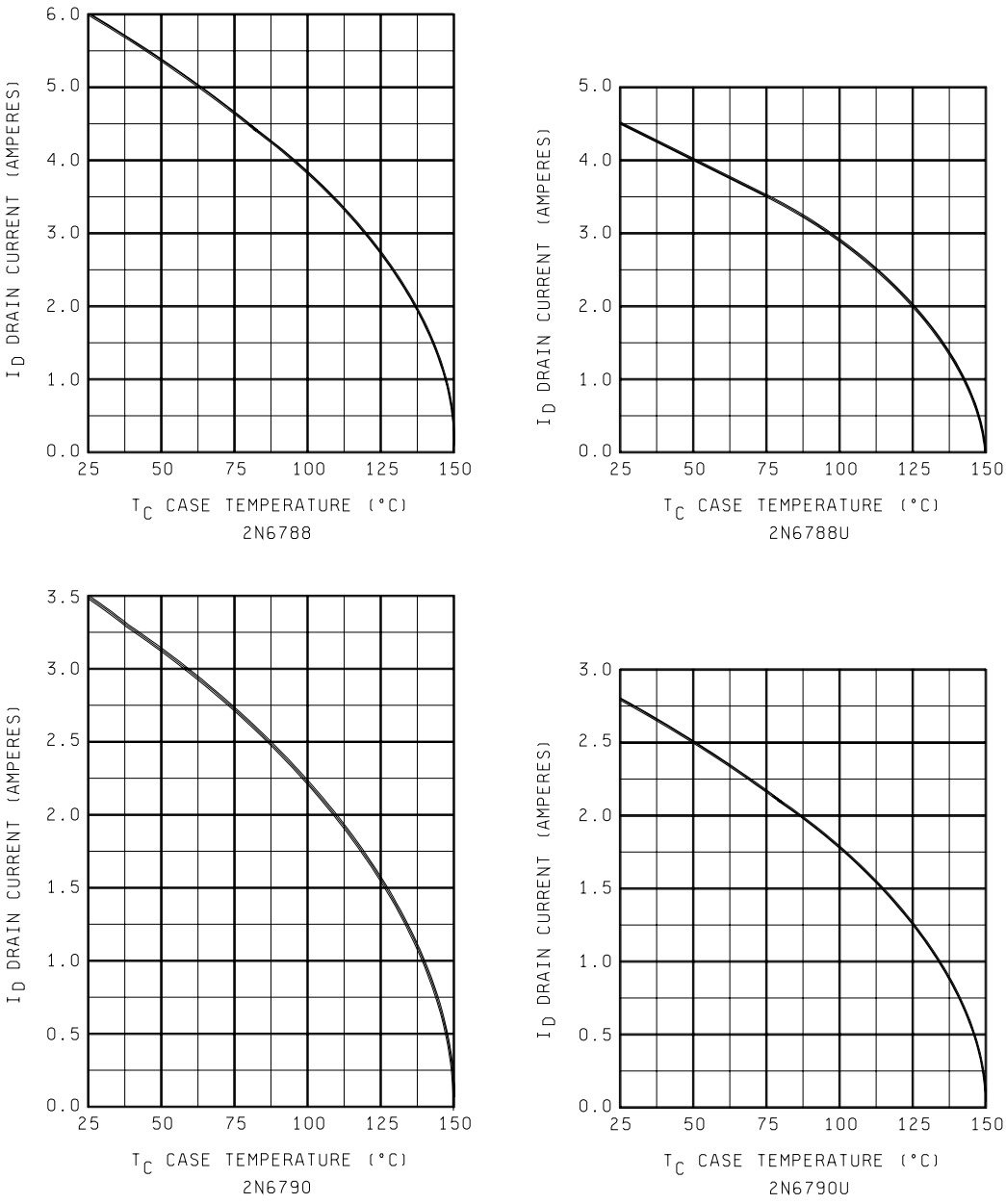


FIGURE 8. Maximum drain current vs case temperature graphs.

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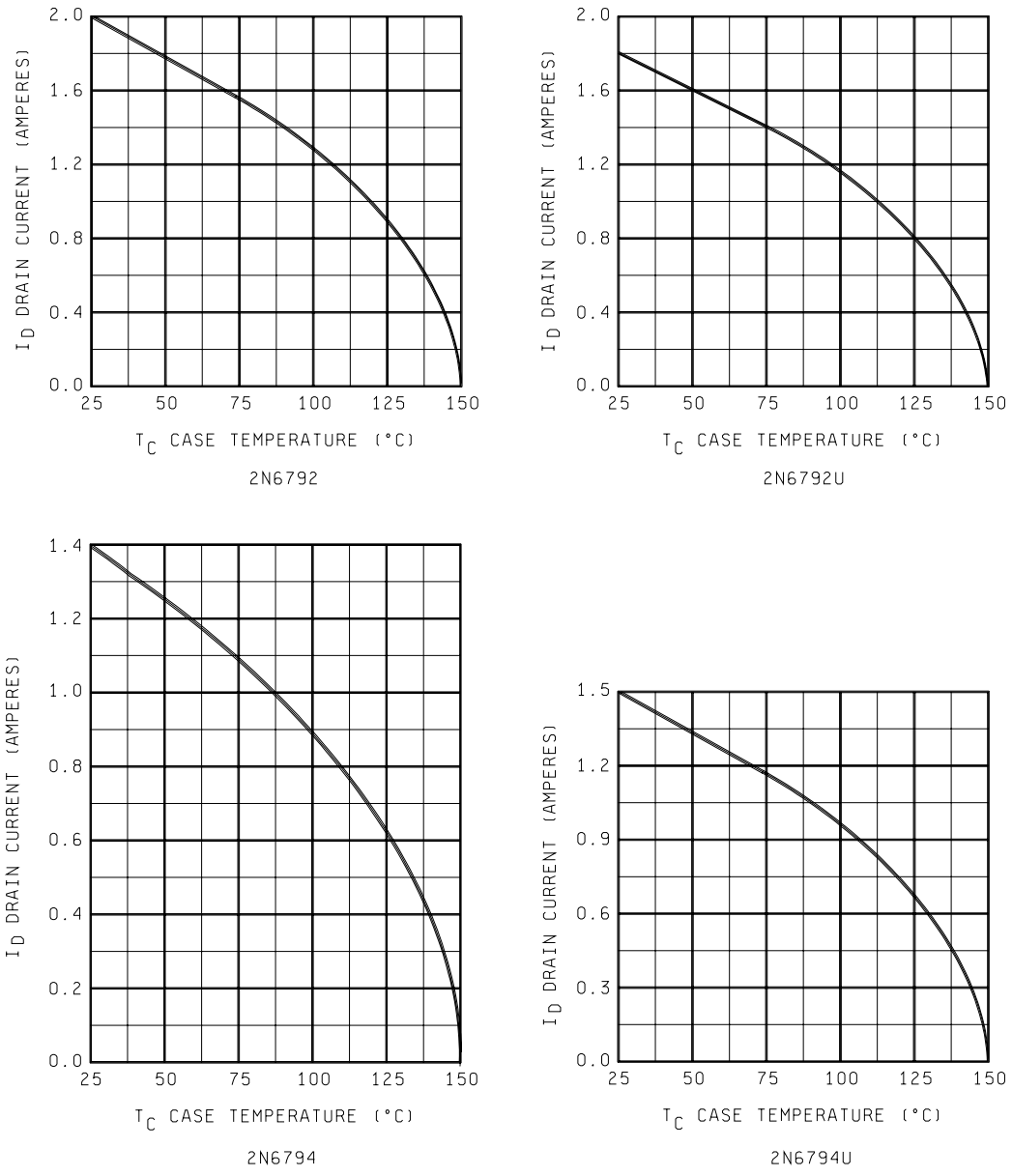
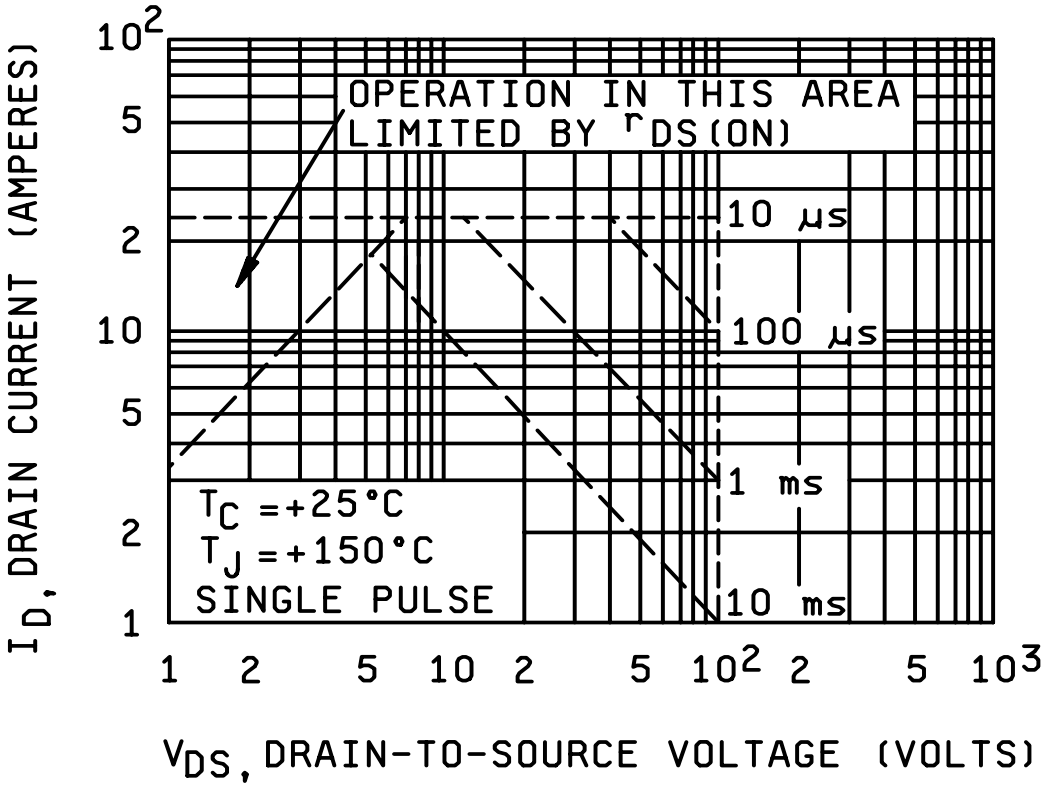


FIGURE 8. Maximum drain current vs case temperature graphs - Continued.

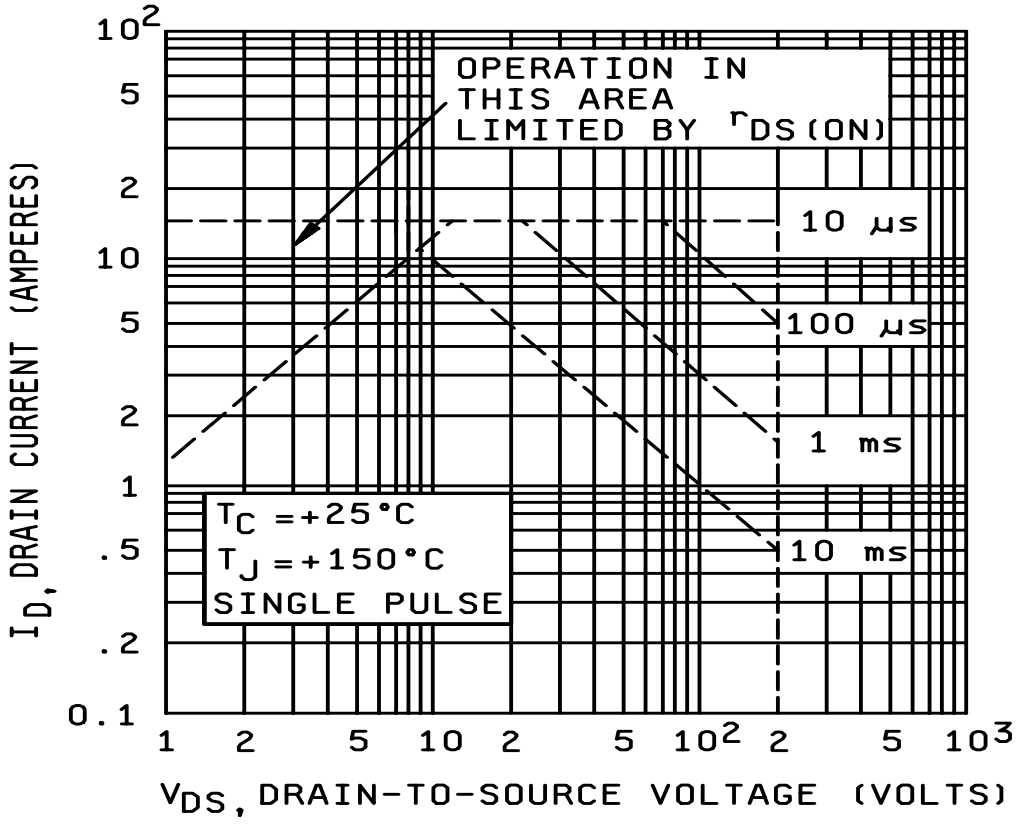
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2N6788, 2N6788U

FIGURE 9. Maximum safe operating area.

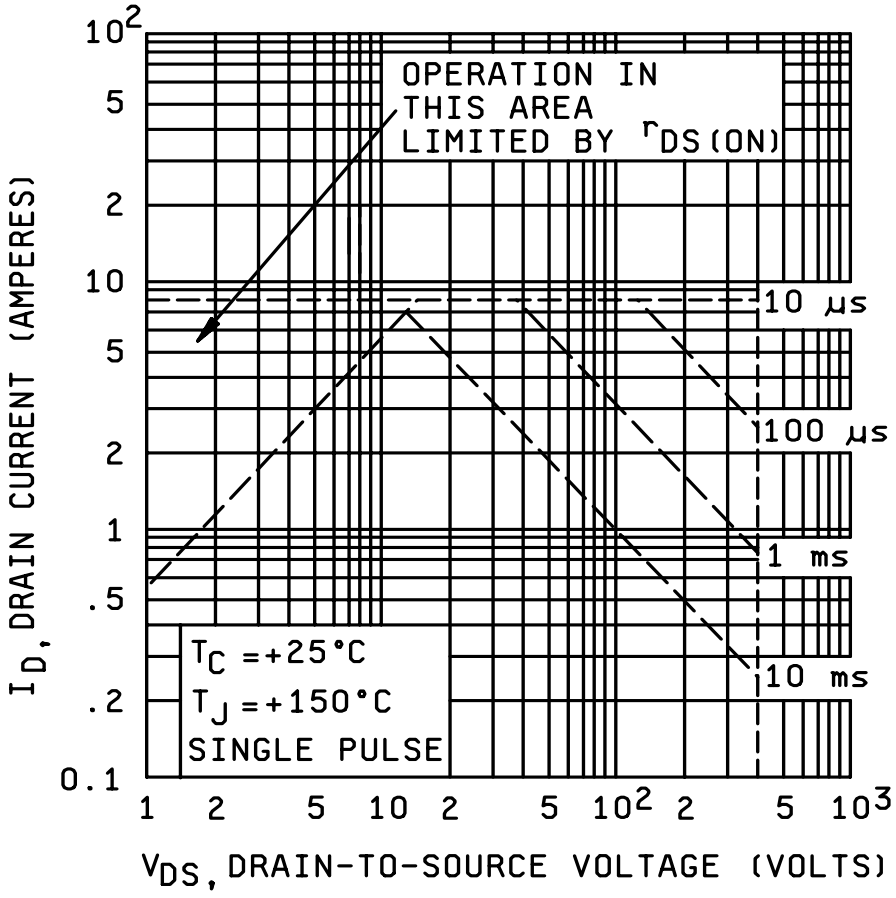
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2N6790, 2N6790U

FIGURE 9. Maximum safe operating area - Continued.

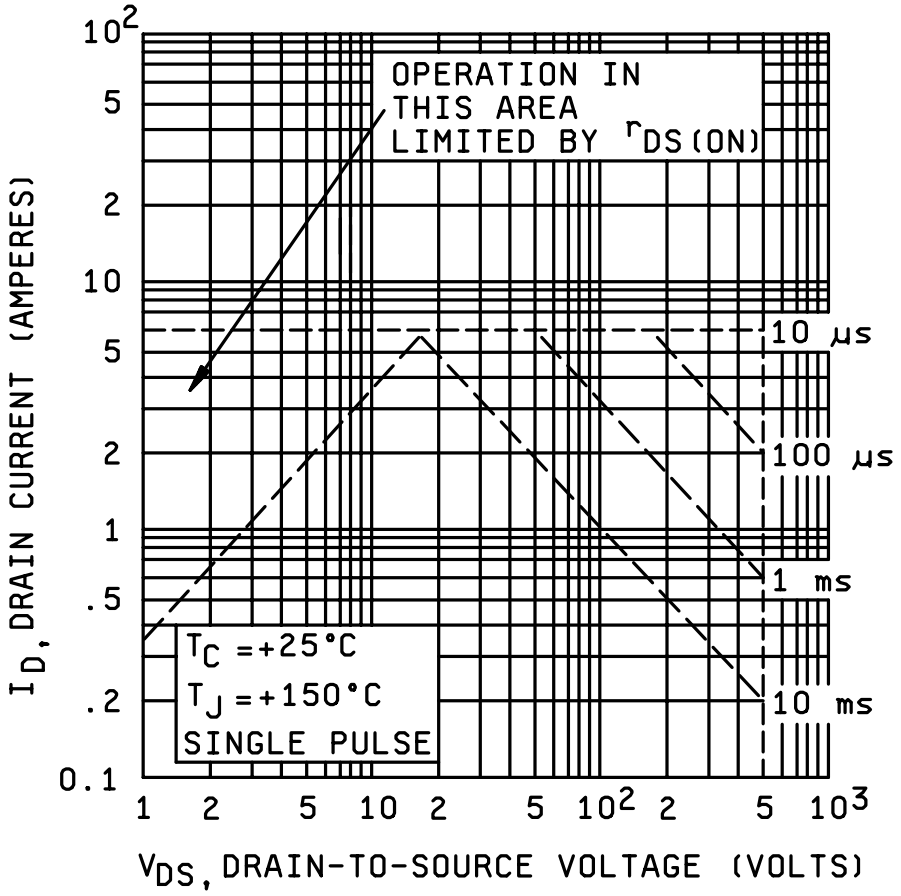
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2N6792, 2N6792U

FIGURE 9. Maximum safe operating area - Continued.

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2N6794, 2N6794U

FIGURE 9. Maximum safe operating area - Continued.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

e. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 3, 4, 5, and 6).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Cross-reference complement list. Parts from this specification may be used to supersede the following commercial Part or Identifying Number (PIN) listed below. Complementary transistors are covered by MIL-PRF-19500/564:

Preferred types (1)	Commercial types
2N6788	IRFF120
2N6790	IRFF220
2N6792	IRFF320
2N6794	IRFF420
2N6788U	IRFE120
2N6790U	IRFE220
2N6792U	IRFE320
2N6794U	IRFE420

(1) Prefixes are JAN, JANTX, JANTXV, or JANS

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6.5 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example, JANHCA2N6788) will be identified on the QML.

JANC ordering information			
PIN	Manufacturer		
	59993	18722	17856
2N6788	JANHCA2N6788 JANKCA2N6788	JANHCB2N6788 JANKCB2N6788	JANHCC2N6788 JANKCC2N6788
2N6790	JANHCA2N6790 JANKCA2N6790	JANHCB2N6790 JANKCB2N6790	JANHCC2N6790 JANKCC2N6790
2N6792	JANHCA2N6792 JANKCA2N6792	JANHCB2N6792 JANKCB2N6792	JANHCD2N6792 JANKCD2N6792
2N6794	JANHCA2N6794 JANKCA2N6794	JANHCB2N6794 JANKCB2N6794	JANHCD2N6794 JANKCD2N6794

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19, 71, 99

Preparing activity:
DLA - CC

(Project 5961-2007-077)

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